① Let A be an n-element set and B be an m-element set. Find the number of injections of A into B in each of the following cases.

(i) $n=m$ (ii) $n>m$ (iii) $n<m$.

② Define $f: \mathbb{N}\times\mathbb{N} \longrightarrow \mathbb{N}$ by $f(a,b) = 2^{a-1}(2b-1)$. Then prove that f is a bijection and find f^{-1}.

③ Let X be a non-empty set. Prove that $fog = gof$ for all bijections f and g of X onto itself if and only if X has atmost two elements in X.

④ Which of the following are equivalence relations?

(i) $\{(a,b) \in \mathbb{R}\times\mathbb{R} : a-b \text{ is a rational number}\}$

(ii) $\{(a,b) \in \mathbb{Q}\times\mathbb{Q} : a-b \in \mathbb{Z}\}$

(iii) Let $\mathbb{Z}^* = \mathbb{Z}\setminus\{0\}$ and

$$R = \{(a,b) \in \mathbb{Z}^*\times\mathbb{Z}^* : b = 2^n a \text{ for some } n \in \mathbb{Z}\}.$$

(iv) $\{((a,b),(c,d)) \in \mathbb{R}^2\times\mathbb{R}^2 : a^2+b^2 = c^2+d^2\}$

(v) $\{(a,b) \in \mathbb{N}\times\mathbb{N} : n \text{ divides both } a \text{ and } b \text{ for some } 1<n\in\mathbb{N}\}$.

(vi) $\{(a,b) \in \mathbb{R}\times\mathbb{R} : ab \text{ is a rational number}\}$

(vii) Let $\mathbb{R}^* = \mathbb{R}\setminus\{0\}$; and

$$R = \{(a,b) \in \mathbb{R}^*\times\mathbb{R}^* : 0 < ab^{-1} \in \mathbb{Q}\}.$$

(5) Let $X = \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$ and $R = \{((a,b),(c,d)) \in X \times X : ad = bc\}$. Prove that R is an equivalence relation on X.

(6) Let $X = \mathbb{N} \times \mathbb{N}$ and $R = \{((a,b),(c,d)) \in X \times X : a+d = b+c\}$. Prove that R is an equivalence relation on X.

(7) Describe the equivalence classes corresponding to each of the equivalence relations given in exercise (5) and (6) above.

(8) Find all equivalence classes of the following equivalence relations.

(i) $\{(a,b) \in \mathbb{R} \times \mathbb{R} : a - b \in \mathbb{Q}\}$

(ii) $\{(a,b) \in \mathbb{Z} \times \mathbb{Z} : a - b$ is a multiple of n for $n \in \mathbb{N}\}$.

(9) Let $f : X \to Y$ and $g : Y \to X$ be functions such that gof is an injection and fog is a surjection. Then prove that both fog and gof are bijections.

(10) Let X be any set and $2 = \{0,1\}$. Prove that the map $\chi : P(X) \longrightarrow 2^X$ defined by

$$\chi(A)(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$$

for any $A \in P(X)$ and $x \in X$, is a bijection.

(Here 2^X denote the set of all functions of X into $\{0,1\}$.)